

MATH 458: Scientific Computing II

Exam Number: 20892768

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1 Purpose

This report covers the MPI parallel implementation of vector-vector multiplication. The program distributes the vectors among MPI processes, computes a local dot product on each process, and combines the results using MPI reduction. The execution time was recorded for $N = 2.5 \times 10^8$ using 1 to 8 cores.

2 Implementation

2.1 MPI distribution

Rank 0 generates the two vectors and distributes their portions using `MPI_Scatterv`. The data is divided between the processes according to the number of available ranks, with the remainder distributed among the first ranks.

Each process computes its local vector-vector product using `vector_vector`. The local results are then combined at rank 0 using `MPI_Reduce` with the `MPI_SUM` operation.

2.2 Timing

The computation is timed using `MPI_Wtime`. The timed region includes the distribution of the vectors, the local vector-vector multiplication, and the reduction of the partial results.

3 Results

Table 1 shows the measured execution times for $N = 2.5 \times 10^8$.

The speed-up is calculated relative to the one-core run:

$$S_p = \frac{T_1}{T_p}.$$

4 Strong Scaling

Figure 1 shows the measured execution time as the number of cores is increased.

The execution time decreases from 2.220096 s with one core to 1.864627 s with three cores. The three-core run gives the best measured speed-up of approximately 1.19. From four to eight cores, the execution time increases, giving speed-up values below one.

Table 1: Measured parallel execution times and speed-up.

Cores	V.V time (s)	Speed-up
1	2.220096	1.0000
2	1.969520	1.1272
3	1.864627	1.1906
4	9.935226	0.2235
5	14.763999	0.1504
6	18.037921	0.1231
7	20.334098	0.1092
8	22.076425	0.1005

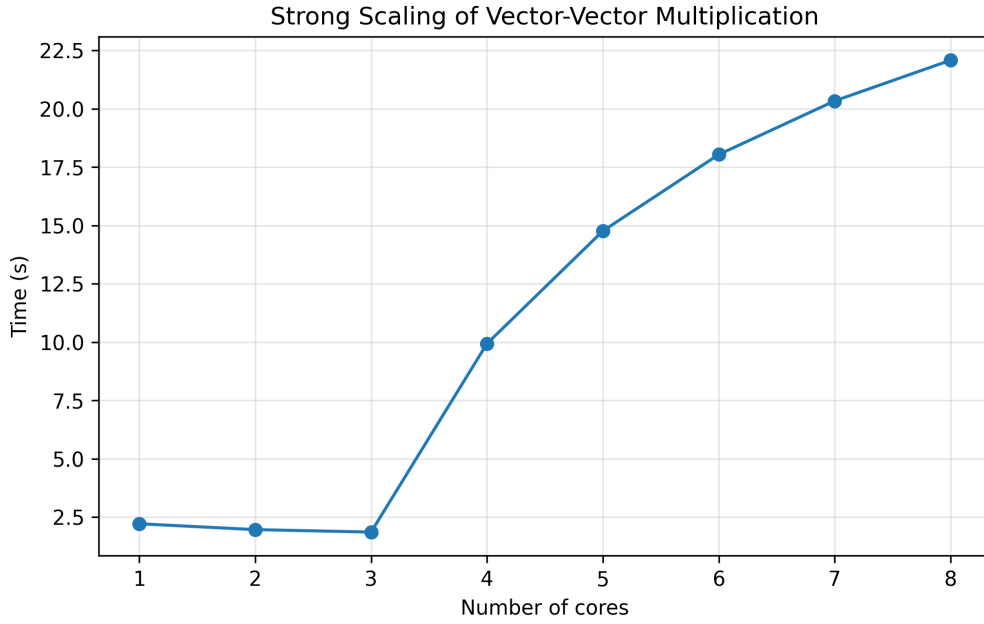


Figure 1: Strong scaling of vector-vector multiplication for $N = 2.5 \times 10^8$.

5 Conclusion

The MPI implementation achieves a small speed-up up to three cores. However, the measured performance decreases when more than three cores are used. For the given test, the best measured performance is therefore obtained with three cores.